⚡ 100% ONCHAIN • PEER-TO-PEER • ZERO HOUSE EDGE

PotShot

"One throw. One pot."

Trustless 1v1 Rock-Paper-Scissors wagering on the BOT Testnet. Players secretly commit with cryptographic hashes, match exact native BOT stakes, and reveal in real time. Winner takes the whole pot—with automated timeout forfeiture protection and zero platform take.

NETWORK	CONTRACT ADDRESS	SECURITY	STATUS
BOT Testnet (968)	0x5A68...834e	OZ 5.x ReentrancyGuard	● Live & Verified

The Problem With Modern Onchain Gaming

Predatory house cuts, mempool front-running, and dishonest player abandonment

MARKET INEFFICIENCIES



1. Predatory House Edges & Casino Bias

Traditional online casinos and Web3 gambling apps impose 2%–10% house edges, ensuring players are mathematically destined to lose over time. Players crave fair, skill-and-mind-game 50/50 PvP wagers without feeding an intermediary.



2. Mempool Sniping & Front-Running

In naive turn-based onchain games, submitting moves unencrypted allows MEV bots, miners, or watchful opponents to inspect transactions in the mempool and counter the move before block confirmation.



3. The "Losing Player Freeze" / Griefing Bug

In typical two-player reveal games, if Player A reveals a winning move first, Player B realizes they lost and simply abandons the session—leaving the creator's stake frozen indefinitely in smart contract escrow.



4. Fragile Oracles & Vulnerable Push Transfers

Games relying on Chainlink VRF or external keepers suffer from latency and fee overhead. Furthermore, contracts that push ETH/BOT directly to external addresses can be frozen by malicious receiver fallback reverts.

The Solution: Trustless Cryptographic Duels

A mathematically fair 1v1 Rock-Paper-Scissors arena with zero house edge

POTSHOT INNOVATION

🛡️ Cryptographic Commit-Reveal

Moves are permanently hidden until both players lock in. The commitment formula hashes the player's address, choice, and random 32-byte salt:

`keccak256(abi.encode(player, move, salt))`

Nobody can see or front-run the opponent's move.

⏰ Anti-Griefing Timeout Engine

A built-in deadline countdown ensures nobody can hold funds hostage. If a player hides their move after seeing the opponent's reveal, the honest revealer automatically wins the entire 2× pot by forfeit via `claimTimeout()`.

💎 100% P2P & Pull-Payments

Zero fees extracted by the protocol. Every win, draw, or refund credits `pendingBalance`. Funds exit only when the user executes `withdraw()`, preventing external reentrancy and blocking malicious revert griefing.

0%	1v1	100%	2×
PROTOCOL TAKE RATE	PVP SKILL & ODDS	FULLY AUTONOMOUS ONCHAIN	WINNER TAKES POT

Four-Step Trustless Game Lifecycle

Deterministic state machine transitions enforce fair play from stake to payout

PROTOCOL ARCHITECTURE

1

Create Game

Player A selects Rock ✊, Paper ✋, or Scissors ✌️, enters BOT stake, generates a random 32-byte salt, and broadcasts `createGame(commitA)`.

STATUS: OPEN

2

Join Game

Player B matches the exact stake with their own secret commitment. `joinGame(id, commitB)` starts the reveal countdown timer (1 hour).

STATUS: COMMITTED

3

Reveal Moves

Both players broadcast `reveal(id, move, salt)`. The contract verifies the hash matches the commitment. The UI animates fists shaking.

SHOWDOWN PHASE

4

Showdown Settle

Standard RPS rules:

- Decisive Win: Winner takes 2× stake.
- Tie: Both refunded.
- Timeout: Non-revealer forfeits full pot to revealer.

STATUS: WON / DRAW

SCENARIO	PLAYER A MOVE	PLAYER B MOVE	RESOLUTION	PAYOUT MECHANISM
Decisive Clash	✊ Rock	✌️ Scissors	**Player A Wins**	2× Stake to Player A pendingBalance
Mutual Draw	✋ Paper	✋ Paper	**Draw**	1× Stake refunded to each player
Attempted Griefing	✊ Revealed	🔒 Never Revealed	**Player B Forfeits**	2× Stake awarded to Player A on claimTimeout

Security & Attack Resistance

Formalized defense against copy-attacks, front-running, reentrancy, and DoS

SECURITY ARCHITECTURE



Anti-Copy Attack Scheme

In naive commitment contracts, Player B can copy Player A's hash without knowing the move. PotShot embeds `player` in the commitment:

A copied hash creates an un-revealable commitment for Player B, resulting in guaranteed forfeiture.



Reentrancy & DoS Immunity

External calls are never made during game resolution. Funds are only transferred during an explicit `withdraw()` execution guarded by OpenZeppelin's `ReentrancyGuard`. A malicious contract that reverts on ETH receipt only harms itself.



Zero Admin Risk

No owner, no upgrade proxy, no pause switch, and no fee treasury. The contract cannot be modified, seized, or drained. The immutable code deployed to block 29,637,423 is perpetual and fully autonomous.

Game-Theoretic Dominant Strategy: Honest Reveal

Under traditional escrow, a player who realizes their move lost has an incentive to stall. Under PotShot, failing to reveal forfeits 100% of the pot to the opponent. Revealing always maintains an expected payoff ≥ forfeiting, making immediate reveal the strictly dominant strategy for all players.

Rigorous Testing & Invariant Verification

32 passing Foundry test suites covering fuzzing, edge cases, and state invariants

QUALITY ASSURANCE

32 / 32

Test Suites Passed

Zero failures, zero warnings. Full coverage of all 9 move combinations, double-reveal blocks, constructor bounds (1 min to 7 days), and pagination.

256

Fuzzing Runs

Randomized stakes, player salts, and execution orders prove native BOT is strictly conserved: never minted, burned, or leaked under any permutation.

4,096

Invariant Transitions

Stateful handler executing across 64 runs proves: `contract.balance == sum(Open) + 2*sum(Committed) + sum(pending)` holds true across every possible state.

```
[PASS] testFuzz_GamePlayConservesBOTAndAwardsCorrectWinner(uint96,uint8,uint8,bytes32,bytes32,bool) (runs: 256)
[PASS] test_All9MoveCombinations_BothRevealOrders() (all 3 draws, 6 decisive victories)
[PASS] test_CopyAttack_Prevented() (player address bound in hash)
[PASS] test_HidingLosingMove_Fails() (revealer awarded 100% pot on forfeit)
[PASS] invariant_BalanceMatchesLockedStakesPlusPendingBalances() (runs: 64, calls: 4096, reverts: 0)
```

High-Energy Retro-Arcade Frontend

Vite + React + TypeScript + viem with client-side secret backup and live showdowns

UI & UX POLISH

🕹️

Chunky Neon Arcade Aesthetic

Vibrant cyberpunk arcade theme with Hot Magenta and Electric Cyan accents, 3D offset drop-shadow buttons, animated fist-shaking showdowns, and live countdown timers.

💾

Client-Side Secret Resilience

Salts generated with cryptographic randomness (`crypto.getRandomValues`) and stored in `localStorage`. Includes one-click JSON backup export and manual fallback recovery.

⚡

Shareable Hash Routing

Direct peer-to-peer challenge links via hash routing (`#/game/:id`). Allows players to post duel links in Discord or Telegram for instant head-to-head showdowns.

💰 Instant Pull-Payment Banner

When winnings or refunds are credited to `pendingBalance`, a luminous yellow banner notifies the player with an instant "Withdraw All" button.

📊 Live Player Statistics

Direct onchain tracking of total wins 🏆, losses 💀, and draws 🤝 displayed in real time for connected wallets.

Ecosystem Synergy & Growth Roadmap

Driving native BOT transaction velocity through casual competitive gaming

STRATEGIC VISION

PHASE 1 • COMPLETED

Testnet Protocol

Deployed on BOT Testnet (Chain ID 968). Fully tested contract, zero linter warnings, and retro-arcade frontend with secret backup.

PHASE 2 • Q1 2027

Mainnet Launch

Deploy to BOT Mainnet (Chain ID 677) with optimized RPC clusters and community wagering pools.

PHASE 3 • Q2 2027

Telegram Mini App

Launch seamless Telegram and Discord bots allowing Web3 community members to challenge each other in chat groups instantly.

PHASE 4 • Q3 2027

Tournament SDK

Expand into Best-of-3 / Best-of-5 tournament brackets and release the PotShot SDK for third-party games to integrate commit-reveal showdowns.

Why the BOT Blockchain is the Ideal Home:

High-frequency casual games require fast block times and low transaction fees. The BOT Network's sub-second finality allows players to create, join, reveal, and settle duels in seconds, driving organic token velocity and wallet adoption without artificial incentives.

Onchain Coordinates & Summary

Fully verified, deployed, and playable on the BOT Testnet

VERIFIED DEPLOYMENT

Deployment Coordinates

Contract Address:
`0x5A682a3450a31fad39F25bB3bCEcF2bDBFdf834e`

Deployment Tx:
`0x7dd2baac2bc0ede33e99a6a586f41a2fa722fb4f5e2549c09bd83330ba132c14`

Network / Chain ID: **BOT Testnet (968)**

RPC Endpoint: https://rpc.bohr.life

BohrScan Explorer: https://scan.bohr.life

Why PotShot Wins

- **Zero Platform Edge:** 100% of wagers flow directly to players.
- **Cryptographic Secrecy:** No miner or peer front-running.
- **Grief-Proof Timeouts:** Non-revealing opponents forfeit the full pot.
- **Isolated Payouts:** Pull-based withdrawals block reentrancy exploits.
- **Retro Polish:** Engaging, high-energy arcade interface designed for casual gamers.

"One throw. One pot."

Play live now at `http://localhost:5173/` or inspect the contract on BohrScan.